2012-2013 – Master 2 – Macro I

Lecture notes #9 : the Mortensen-Pissarides matching model

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Changes from version 1.0 are in red
Changes from version 1.1 are in purple

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1. A simple framework

The matching function

- The basic building block is the matching function, which relates hirings per unit of time to the two key inputs in the search process, unemployment and vacancies :

$$H_t = m(U_t, V_t). \tag{1}$$

- Here $H_t =$ the gross hiring rate per unit of time, $U_t =$ the number of unemployed workers, $V_t =$ the number of vacant jobs.
- The matching function is similar to a production function, and we assume it has the same properties.
- Note that in this framework, unemployment and vacancies are not a waste : they are a productive input in the production of new matches.
- This defines the process for job creation. To begin with, we assume a simple process of job destruction : a fraction s of all jobs is destroyed per unit of time.

1. A simple framework

Evolution of unemployment

- Let $\bar{L}$ = the total labor force, L_t = employment at t. Then we can define the hiring, unemployment, and vacancy rates in relation to the total workforce :

$$\begin{aligned} u_t &= \frac{U_t}{\bar{L}} = \frac{\bar{L} - L_t}{\bar{L}}, \\ v_t &= \frac{V_t}{\bar{L}}, \\ h_t &= \frac{H_t}{\bar{L}}. \end{aligned}$$

- Because of constant returns to scale, we can rewrite (1) as

$$h_t = m(u_t, v_t).$$

- The evolution of the unemployment rate is

$$\begin{aligned} \frac{du}{dt} &= -h_t + s(1 - u_t) \\ &= -m(u_t, v_t) + s(1 - u_t). \end{aligned}$$

1. A simple framework

The Beveridge curve

- This defines a $du/dt = 0$ locus in the (u, v) plane which is called the "Beveridge curve".
- Along this locus, we have

$$\begin{aligned} 0 &= -m'_u du - m'_v dv - sdu \\ \implies & \frac{dv}{du} = -\frac{m'_u + s}{m'_v} < 0. \end{aligned}$$

- Furthermore,

$$\begin{aligned} \frac{d^2 v}{du^2} &= -\frac{\left(m''_{uu} + m''_{uv}\frac{dv}{du}\right) m'_v - (m'_u + s)(m''_{uv} + m''_{vv}\frac{dv}{du})}{m'^2_v} \\ &\propto (-m''_{uu} m'_v) + (m''_{vv}\frac{dv}{du})(m'_u + s) + (m''_{uv}(m'_u + s - \frac{dv}{du} m'_v)), \end{aligned}$$

  and all the terms in parentheses in the last expression are > 0, therefore

$$\frac{d^2 v}{du^2} > 0.$$

- Note : "$\propto$" means "proportional with the same sign".

1. A simple framework

Labor market tightness

- Convexity of the Beveridge curve : because decreasing marginal returns to each input in the matching function.
- When I increase vacancies by one unit when vacancies are large, the effect on hirings is small, and only a small reduction in unemployment would maintain a balance between employment outflows and inflows.
- Given constant returns, it is easier to think in terms of labor market tightness rather than vacancies. By definition, labor market tightness is

$$\theta = v/u.$$

- The probability per unit of time of finding a job is

$$p = h/u = \frac{m(u,v)}{u} = m(1,\theta) = p(\theta), p' > 0, p'' < 0.$$

- The probability per unit of time of filling a vacancy is

$$q = \frac{m(u,v)}{v} = m(\frac{1}{\theta},1) = q(\theta), q' < 0.$$

1. A simple framework

Along the Beveridge curve

- Furthermore,

$$p(\theta) = \theta m(\frac{1}{\theta}, 1) = \theta q(\theta).$$

- The Beveridge curve can be reexpressed in the (u, θ) plane :

$$\begin{aligned} \dot{u} &= s(1-u) - up(\theta) \\ &= s(1-u) - u\theta q(\theta). \end{aligned}$$

- Along this curve :

$$\begin{aligned} \frac{d\theta}{du} &= -\frac{s+p}{up'} < 0; \\ \frac{d^2\theta}{du^2} &\propto -up'^2\frac{d\theta}{du} + (s+p)(p' + p''u\frac{d\theta}{du}) > 0. \end{aligned}$$

- The Beveridge curve delivers one dynamic relationship between u and v (or θ). Above it vacancies are larger than in steady state, so unemployment is falling. Below it, unemployment is rising. Hence the arrows on Figure 1.
- To complete the model we need another relationship between u and θ. This will come from labor demand.

1. A simple framework

The Beveridge curve

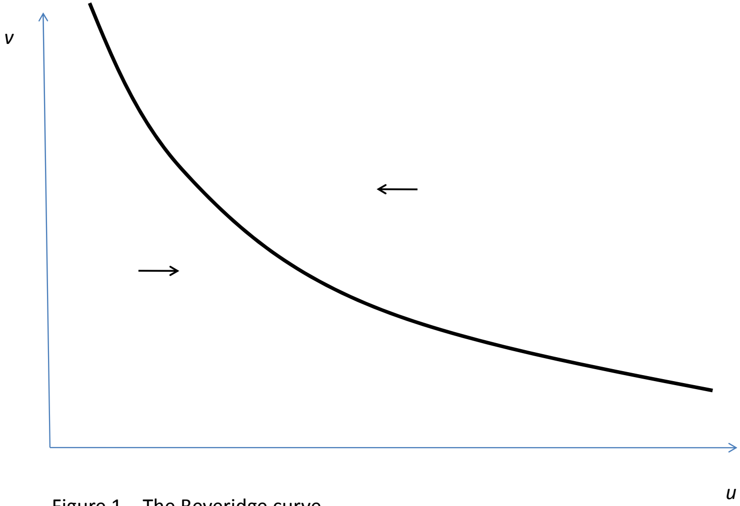

Figure 1 – The Beveridge curve

1. A simple framework

Labor demand

- We assume there is a single homogeneous good. Once a worker finds a job, he produces a constant flow of this good equal to y per unit of time. He is paid a fixed wage w. There is a fixed real interest rate equal to r. Let J_t be the value of the firm at t. Since the job is destroyed with flow probability s, the asset valuation equation for J is
$$rJ = y - w + \dot{J} - sJ. \tag{2}$$
- The only non explosive solution is
$$J = \frac{y - w}{r + s}.$$
- To recruit workers, firms must post vacancies. Posting a vacancy costs c per unit of time. Let V_v be the value of a vacancy. Its asset valuation equation is
$$rV_v = -c + q(\theta)(J - V_v) + \dot{V}_v.$$

1. A simple framework

Labor demand (continued)

- There is free entry in posting vacancies. Therefore,

$$V_v = 0.$$

- Thus we get

$$J = \frac{c}{q(\theta)}.$$

- Note that the expected duration of a vacancy is $1/q(\theta)$, therefore this tells us that the value of a job is equal to the average recruiting cost per job.
- This determines the equilibrium value of θ, which is constant and equal to

$$\theta = q^{-1}\left[\frac{c(r+s)}{y-w}\right].$$

- Figure 2 shows the adjustment dynamics.

1. A simple framework

Equilibrium

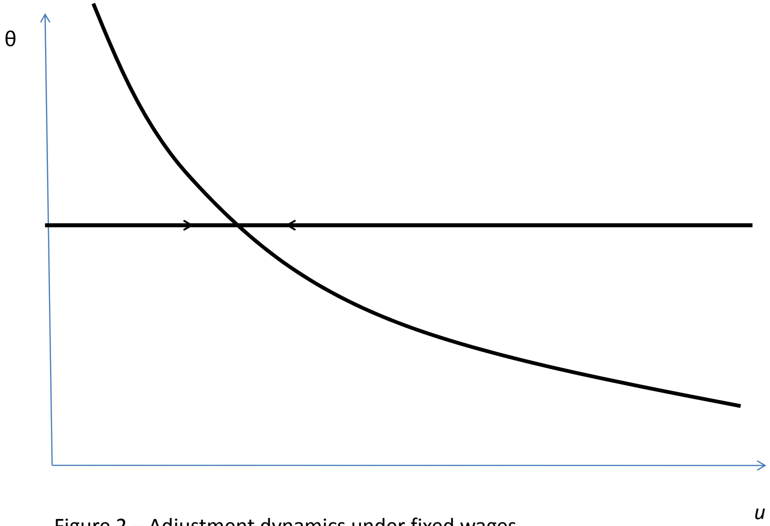

Figure 2 – Adjustment dynamics under fixed wages

1. A simple framework

Comparative statics

- θ goes up, and u falls, if the profitability of a job goes up, i.e. if r goes down, y goes up,w goes down.
- θ goes up if the cost of a vacancy falls.
- All these changes do not affect the BC. Thus the economy moves along the BC. (Figure 3)
- A rise in s shifts both the labor demand curve and the BC through the discounting and mechanical effects of job destruction. (Figure 4)
- Assume shocks to y alternate : this suggests that business cycles induce counter-clockwise loops around the Beveridge curve.

1. A simple framework

Comparative statics

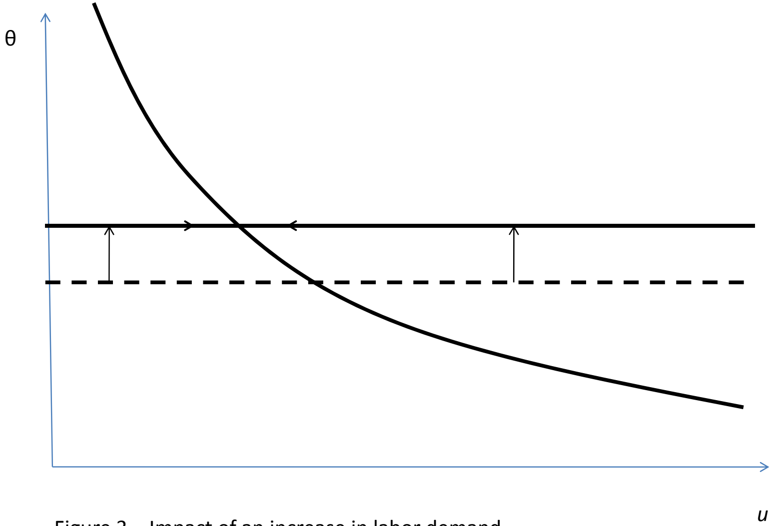

Figure 3 – Impact of an increase in labor demand

1. A simple framework

Comparative statics

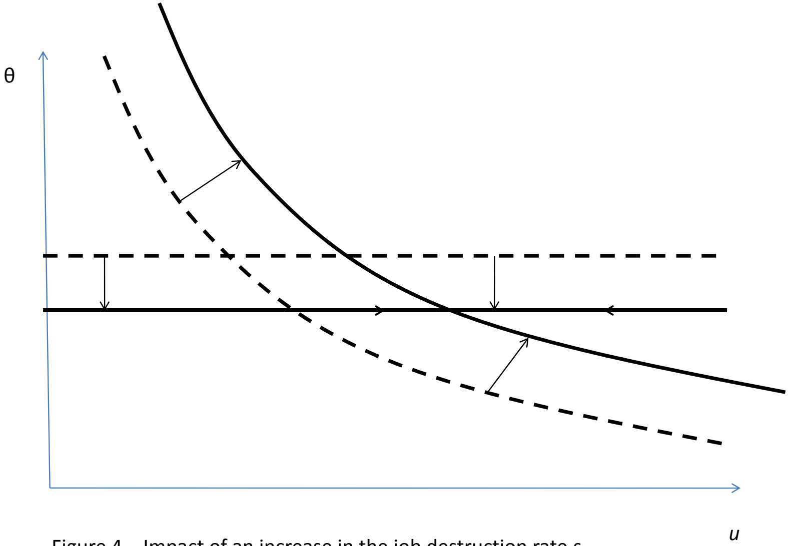

Figure 4 – Impact of an increase in the job destruction rate *s*

2. Endogenous wages

Bargaining

- An elegant way to endogenize wages is to assume that incumbent employees and employers bargain over the surplus created by the sunk recruiting costs.
- Bargaining is individual between each worker and the firm. It is easiest to assume that 1 firm = 1 job.
- Let V_e be the value of being employed, V_u be the value of being unemployed.
- The asset valuation equations for V_e and V_u are (assuming no unemployment benefit)

$$\begin{aligned} rV_e &= w + s(V_u - V_e) + \dot{V}_e; & (3) \\ rV_u &= \theta q(\theta)(V_e - V_u) + \dot{V}_u. & (4) \end{aligned}$$

2. Endogenous wages

Bargaining (continued)

- It is convenient to use the net surplus of the match, i.e.

$$W = J + V_e - V_u.$$

- Consolidating (2),(3), and (4) we get

$$rW = y - \theta q(\theta)\varphi W - sW + \dot{W}. \tag{5}$$

2. Endogenous wages

Bargaining (continued)

- At each date wages are set so as to maximize the joint log Nash product :

$$\ln\left[(J - V_v)^{1-\varphi}(V_e - V_u)^{\varphi}\right].$$

- Bargaining takes place at the firm-worker level, taking as given aggregate conditions V_u and V_v.
- This is because the bargaining sets the wage *within* th match, not for all the economy
- Recall that $J = \frac{y-w}{r+s}$
- Recall that $rV_e = w + s(V_u - V_e) + \dot{V}_e$ so that the non-explosive solution is $V_e = \frac{w+sV_u}{r+s}$

2. Endogenous wages

Bargaining (continued)

- For any increase in wages Δw we have (all else equal) $\Delta V_e = \frac{1}{r+s}\Delta w$ and $\Delta J = -\frac{1}{r+s}\Delta w = -\Delta V_e$.
- Therefore the FOC is :

$$\frac{1-\varphi}{J - V_v} = \frac{\varphi}{V_e - V_u}.$$

- Since $V_v = 0$, this is equivalent to

$$V_e = V_u + \frac{\varphi}{1-\varphi} J. \tag{6}$$

- That is :

Value of being employed = Outside option + rent.

2. Endogenous wages

Bargaining (continued)

- Since $J = \frac{c}{q(\theta)}$, the rent is proportional to the total recruiting cost that has been spent.
- Equation (6) determines the wage despite that the wage does not explicitly appear in it.
- We now need to derive a relationship between θ and u in this more complicated model.

- Note that (6) implies that this net surplus is shared in proportion $(\varphi, 1-\varphi)$, that is

$$\begin{aligned} V_e - V_u &= \varphi W, \\ J - V_v &= J = (1-\varphi) W. \end{aligned}$$

2. Endogenous wages

The real wage

$$rW = y - \theta q(\theta)\varphi W - sW + \dot{W}. \qquad (5)$$

- The term in $\theta q(\theta)\varphi W$ is the opportunity cost to the worker of being employed in this match instead of being looking for another job which would yield a net value φW to the worker and have an arrival rate $\theta q(\theta)$.
- Last, W can be expressed as a function of θ, since

$$\frac{c}{q(\theta)} = J = W(1-\varphi).$$

- We get a dynamic equation for θ :

$$\frac{(r+s)c}{(1-\varphi)q(\theta)} = y - \frac{c}{(1-\varphi)q(\theta)^2} q'(\theta)\dot{\theta} - \frac{\theta\varphi c}{1-\varphi}. \qquad (7)$$

2. Endogenous wages

Dynamics

- The $\dot{\theta} = 0$ schedule is such that θ is constant and solves

$$\frac{(r+s)c}{(1-\varphi)q(\theta)} = y - \frac{\theta\varphi c}{1-\varphi}.$$

- Since $q' < 0$, (7) has unstable dynamics locally around $\dot{\theta} = 0$.
- As θ is a non-predetermined variable, we pick the only non-explosive solution, i.e. the saddle-path which is horizontal (Figure 5).
- The adjustment dynamics are qualitatively the same as in the case where w is fixed.

2. Endogenous wages

Dynamics

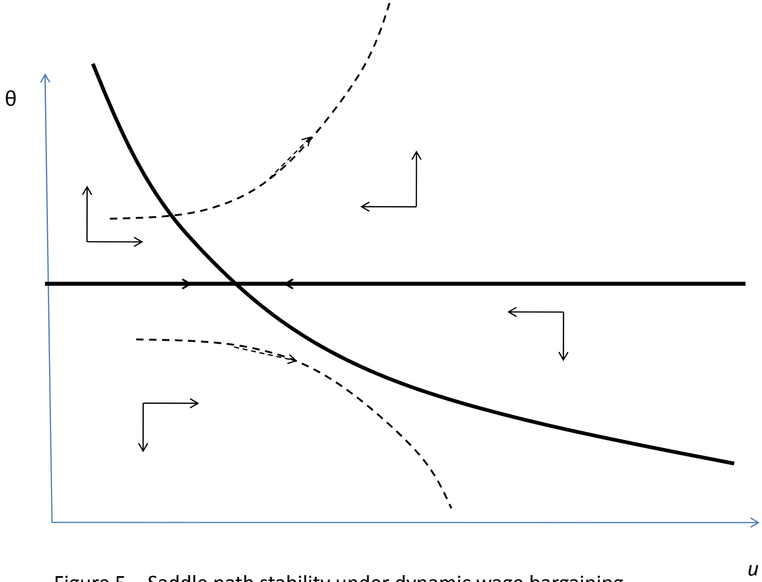

Figure 5 – Saddle path stability under dynamic wage bargaining

3. Endogenous job destruction

Productivity shocks

- Another direction in which we may want to enrich the model is by endogenizing job destruction.
- For this we assume the firm has idiosyncratic productivity shocks.
- Productivity at any date is $y = \sigma\varepsilon$, where ε is distributed over $[\varepsilon_l, \varepsilon_u]$.
- Newly created jobs have productivity ε_u
- With arrival rate λ per unit of time, ε is then redrawn with a c.d.f. $F(), F' = f$, over $[\varepsilon_l, \varepsilon_u]$.
- The endogenous job destruction margin is determined by a threshold ε_d such that the job is destroyed if $\varepsilon < \varepsilon_d$.
- Note : it may be that $\varepsilon_d \leq \varepsilon_u$ in which case the job is never destroyed).

3. Endogenous job destruction
Job destruction rate

- The job destruction rate is now
$$s = \lambda F(\varepsilon_d).$$
- The negotiated wage now generally depends on the current value of ε :
$$w = w(\varepsilon).$$
- We need to rewrite the asset valuation equation, in steady state, for J :
$$rJ(\varepsilon) = \sigma\varepsilon - w(\varepsilon) + \lambda \int_{\varepsilon_d}^{\varepsilon_u} (J(x) - J(\varepsilon)) f(x) dx + \lambda F(\varepsilon_d)(0 - J(\varepsilon)). \tag{8}$$

3. Endogenous job destruction

Bargaining

- Similarly, the value of the worker is given by

$$rV_e(\varepsilon) = w(\varepsilon) + \lambda \int_{\varepsilon_d}^{\varepsilon_u} (V_e(x) - V_e(\varepsilon)) f(x) dx + \lambda F(\varepsilon_d)(V_u - V_e(\varepsilon)). \tag{9}$$

- Finally, the value of being unemployed obeys

$$rV_u = \theta q(\theta)(V_e(\varepsilon_u) - V_u). \tag{10}$$

- Let

$$W(\varepsilon) = J(\varepsilon) + V_e(\varepsilon) - V_u.$$

- The Nash bargaining solution implies that

$$\begin{aligned} J(\varepsilon) &= (1-\varphi)W(\varepsilon); \\ V_e(\varepsilon) &= V_u + \varphi W(\varepsilon). \end{aligned} \tag{11}$$

3. Endogenous job destruction
The real wage

- Using (8), (9) and (10) we get the new version of (5) :

$$rW(\varepsilon) = \sigma\varepsilon - \theta q(\theta)\varphi W(\varepsilon_u) + \lambda \int_{\varepsilon_d}^{\varepsilon_u} (W(x) - W(\varepsilon)) f(x) dx - \lambda F(\varepsilon_d) W(\varepsilon).$$

- We note that the integral $\bar{W} = \int_{\varepsilon_d}^{\varepsilon_u} W(x) f(x) dx$ is a constant which is independent of the current value of ε.
- Therefore,

$$W(\varepsilon) = \frac{\sigma}{r+\lambda}\varepsilon + \frac{\lambda \bar{W}}{r+\lambda} - \frac{\theta q(\theta)\varphi W(\varepsilon_u)}{r+\lambda}. \qquad (12)$$

3. Endogenous job destruction

Surplus from a match

- Integrating, we get the actual value of $\bar{W}$

$$
\begin{aligned}
\bar{W} &= \int_{\varepsilon_d}^{\varepsilon_u} W(x) f(x) dx \\
&= \int_{\varepsilon_d}^{\varepsilon_u} \frac{\sigma x - \theta q(\theta)\varphi W(\varepsilon_u) + \lambda \bar{W}}{r+\lambda} f(x) dx \\
&= \frac{\sigma}{r+\lambda} \int_{\varepsilon_d}^{\varepsilon_u} x f(x) dx - (1 - F(\varepsilon_d)) \frac{\theta q(\theta)\varphi W(\varepsilon_u)}{r+\lambda} \\
&\quad + (1 - F(\varepsilon_d)) \frac{\lambda}{r+\lambda} \bar{W} \\
\implies \quad & \bar{W} = \frac{\sigma \int_{\varepsilon_d}^{\varepsilon_u} x f(x) dx - (1 - F(\varepsilon_d)) \theta q(\theta) \varphi W(\varepsilon_u)}{r + \lambda F(\varepsilon_d)}.
\end{aligned}
$$

3. Endogenous job destruction
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- Substituting into (12), we get

$$W(\varepsilon) = \frac{\sigma\varepsilon}{r+\lambda} + \frac{\lambda\sigma}{(r+\lambda)(r+\lambda F(\varepsilon_d))} I(\varepsilon_d) - \frac{\theta q(\theta)\varphi W(\varepsilon_u)}{r+\lambda F(\varepsilon_d)}, \tag{13}$$

where

$$I(\varepsilon_d) = \int_{\varepsilon_d}^{\varepsilon_u} x f(x) dx.$$

- The model is closed by deriving a job creation condition and a job destruction condition.
- Each condition gives us a relationship between ε_d, the job destruction margin, and θ, the labor market tightness parameter.

3. Endogenous job destruction

The job destruction condition

- The job destruction condition is

$$J(\varepsilon_d) = 0.$$

- Therefore :

$$\begin{aligned} W(\varepsilon_d) &= 0 \\ &\Longrightarrow V_e(\varepsilon_d) = V_u. \end{aligned}$$

- In this class of models, the separation decision is jointly privately efficient.

3. Endogenous job destruction

The job destruction condition (continued)

- We can then rewrite :

$$W(\varepsilon) = \frac{\sigma}{r+\lambda}(\varepsilon - \varepsilon_d). \tag{14}$$

- This is because we know from (13) that $W()$ is linear with a slope equal to $\frac{\sigma}{r+\lambda}$, while it must satisfy $W(\varepsilon_d) = 0$.
- Substituting into (13) we get

$$\begin{aligned} 0 &= \frac{\sigma\varepsilon_d}{r+\lambda} + \frac{\lambda\sigma}{(r+\lambda)(r+\lambda F(\varepsilon_d))} I(\varepsilon_d) \\ &\quad - \frac{\theta q(\theta)\varphi}{r+\lambda F(\varepsilon_d)} \frac{\sigma}{r+\lambda}(\varepsilon_u - \varepsilon_d) \\ \iff &\quad \varepsilon_d(r+\lambda F(\varepsilon_d)) + \lambda I(\varepsilon_d) - \theta q(\theta)\varphi(\varepsilon_u - \varepsilon_d) = 0. \end{aligned}$$

- This defines a relationship between ε_d and θ.

3. Endogenous job destruction

The job destruction condition (continued)

- Differentiating, we get

$$[r + \theta q(\theta)\varphi + \lambda F(\varepsilon_d)]\, d\varepsilon_d - \left[\varphi(\varepsilon_u - \varepsilon_d)\frac{d}{d\theta}\theta q(\theta)\right] d\theta = 0.$$

- Since $\frac{d}{d\theta}\theta q(\theta) > 0$, both terms in brackets are positive.
- Hence this defines a positive relationship between θ and ε_d.
- When the labor market is tighter, the opportunity cost of work is larger for the worker, because he could find a job starting at the highest productivity level more quickly if he were unemployed. Therefore the productivity threshold below which it is efficient to destroy the job is higher–jobs are destroyed more often.

3. Endogenous job destruction

The job destruction condition (continued)

Other interesting aspects :

- An increase in φ increases ε_d : this is because the worker expects to get more out of future jobs, thus reducing the value of staying in the current job.
- An increase in λ reduces ε_d : as shocks are more frequent, the option value of waiting until a new shock arrives instead of firing the worker right now goes up ; hence I fire less frequently. This option value is the term in $\lambda \int_{\varepsilon_d}^{\varepsilon_u} (J(x) - J(\varepsilon)) f(x) dx$ in (8). It is always positive because I can always decide to fire the worker later, while if I fire him right now I will have to pay again the hiring cost to re-start my business.

3. Endogenous job destruction

The job creation condition

- The job creation condition is

$$J(\varepsilon_u) = \frac{c}{q(\theta)}.$$

- Using (14) and (11), this is equivalent to

$$(1-\varphi)\frac{\sigma}{r+\lambda}(\varepsilon_u - \varepsilon_d) = \frac{c}{q(\theta)}.$$

- Since $q' < 0$, this defines a negative relationship between ε_d and θ.
- When the labor market is tighter, it takes more time to recruit people. This makes it more costly to get rid of incumbent workers. Therefore the productivity threshold where they are fired is lowered.

3. Endogenous job destruction

The job creation condition (continued)

The other comparative statics are similar to those in the basic model :

- An increase in φ reduces θ : firms get a lower share of the surplus out of each job and post fewer vacancies.
- An increase in r reduces θ, since the cost of funds for paying the vacancy cost goes up.
- An increase in λ reduces θ : this is because the first shock reduces productivity (since initial productivity is ε_u), so I want this shock to occur as late as possible.

3. Endogenous job destruction
Equilibrium

- Equilibrium is determined by the intersection of the JC and JD schedules

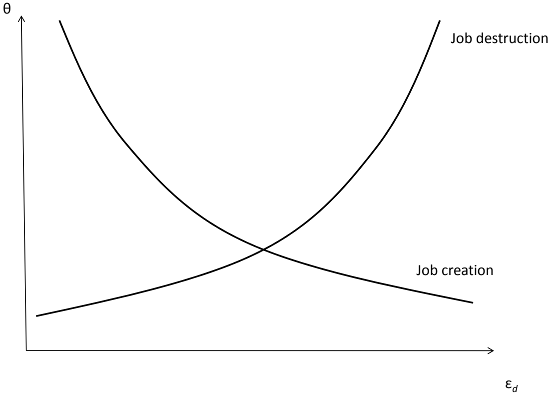

Figure 6 – Equilibrium determination in the Mortensen-Pissarides matching model